

Trainings ▾

General Information

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are due for scheduled maintenance.

Maintenance Procedures at Daily Interval (Section 3)

- Marine Gear Oil - Check¹
- Sea Water Strainer - Clean
- Coolant Level - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Air Cleaner Restriction - Check
- Crankcase Breather Restriction - Check

Maintenance Procedures at 250 Hours or 6 Months (Section 4)

- Zinc Anode - Change²

Maintenance Procedures at 250 Hours or 1 Year (Section 5)

- Lubricating Oil and Filters - Drain³
- Fuel Filter (Spin-On Type) - Change
- Drive Belts - Check

Maintenance Procedures at 500 Hours (Section 6)

- Overhead Set - Adjust⁴

Maintenance Procedures at 1500 Hours or 1 Year (Section 7)

- Sea Water Pump - Repair
- Sea Water Hoses - Check
- Fuel-Water Separator - Change
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check
- Coolant Filter - Change⁵
- Water Pump - Check
- Heat Exchanger Pressure Cap - Inspect for Reuse
- Air Cleaner Assembly (Engine-Mounted) - Clean⁶
- Crankcase Breather Element (Recreational Marine engines) - Change
- Air Leaks, Air Intake and Exhaust Systems - Check
- Turbocharger - Check
- Batteries - Check
- Battery Cables and Connections - Check
- Engine Wiring Harness - Check

- Engine Mounting Bolts - Check
- Marine Vibration Isolator - Check
- Sea Water System - Clean

Maintenance Procedures at 1500 Hours or 3 Years (Section 8)

- Aftercooler Assembly (Sea Water) - Inspect for Reuse⁷

Maintenance Procedures at 6000 Hours or 4 Years (Section 9)

- Cooling System - Flush⁸
- Vibration Damper, Viscous - Check
- Crankcase Breather Element (Recreational Marine engines) - Change

Reference the appropriate sections for complete inspection and maintenance procedures.

1. Follow the gear manufacturer's recommended maintenance procedures for checking, filling, and changing oil. Some gear manufacturers require an initial oil change at low hours, in addition to their regular maintenance.
2. Depending upon the quality of electrical bonding and water conditions, increased maintenance of the zinc anode plugs could possibly be necessary.
3. The lubricating oil and filter change interval is 500 hours if Ultra Low Sulphur Diesel (ULSD) fuel is exclusively used.
4. Engines with rated power under 525hp **only** need the overhead set at 1500 hours.
5. Do **not** change the coolant filter if the SCA concentration level is over three units.
6. For engines equipped with a disposable air cleaner, the air cleaner **must** be replaced at this interval. To determine if the air cleaner is disposable, check the part number on the air cleaner. Refer to Procedure 018-024 in Section V. (</qs3/pubsys2/xml/en/procedures/35/35-018-024-om-mar.html>)
7. Reference the Clean and Inspect for Reuse section in Procedure 010-005 (Aftercooler Assembly (Sea Water)) for more instructions.
8. This cooling system requirement to Flush at this scheduled maintenance includes Drain, Flush, and Fill.

Oil Drain Intervals

QSM11 Marine Engines - Oil Drain Intervals				
Oil Type	Fuel Burned	Hours	Months	Acceptable Filter
CF-4 ⁽¹⁾	6625 liters [1750 gal]	125	6	LF9001 ⁽⁴⁾
CG-4	13,249 liters [3500 gal]	250	6	LF9001 ⁽⁴⁾
CH-4 (CES 20071) ⁽²⁾	15,142 liters [4000 gal]	500 ⁽⁵⁾	12	LF9001 ⁽⁴⁾

QSM11 Marine Engines - Oil Drain Intervals				
Oil Type	Fuel Burned	Hours	Months	Acceptable Filter
CES 20076 ⁽³⁾	15,142 liters [4000 gal]	500 ⁽⁵⁾	12	LF9001 ⁽⁴⁾
Notes: 1. CF-4 oil is not recommended and can only be used if better quality oil is not available. 2. Valvoline™ Premium Blue™ meets and exceeds API CH-4 and CES20071 specifications. 3. Valvoline™ Premium Blue™ 2000 meets and exceeds CES 20076 specifications. 4. Oil filters must meet CES 10065 specifications. LF9001 meets and exceeds CES 10065 specifications. 5. The lubricating oil and filter change interval is 500 hours if Ultra Low Sulphur Diesel (ULSD) fuel is exclusively used.				